

HDOC DAY – 37

Algorithm:

1. Start the program.
2. Display the menu:
 - Convert lowercase string to uppercase
 - Reverse a string
 - Check whether a string is a palindrome
 - Exit
3. Read the user's choice.
4. Use switch-case:
 - Choice 1: Read a string and convert each lowercase character (a-z) to uppercase (A-Z) using ASCII values.
 - Choice 2: Read a string, find its length manually, and print it from the last character to the first.
 - Choice 3: Compare characters from the beginning and end of the string. If all corresponding characters match, it is a palindrome.
 - Choice 4: Exit.
 - Otherwise, display an invalid-choice message.
5. Repeat the menu until the user selects Exit.
6. Stop.

C Program:

```
#include <stdio.h>
```

```
void toUppercase();
```

```
void reverseString();
```

```
void checkPalindrome();
```

```
int main()
```

```
{
```

```
int choice;

do
{
    printf("\n--- STRING OPERATIONS ---\n");
    printf("1. Convert lowercase string to uppercase\n");
    printf("2. Reverse a string\n");
    printf("3. Check if a string is a palindrome\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            toUppercase();
            break;

        case 2:
            reverseString();
            break;

        case 3:
            checkPalindrome();
            break;

        case 4:
```

```

        printf("Exiting program...\n");

        break;

    default:

        printf("Invalid choice! Please enter 1 to 4.\n");
    }

} while (choice != 4);

return 0;
}

/* Function to convert lowercase string to uppercase */
void toUppercase()
{
    char str[100];
    int i;

    printf("Enter a lowercase string: ");
    scanf("%c\n", str);

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }
}

```

```
    printf("Uppercase string: %s\n", str);  
}
```

```
/* Function to reverse a string */
```

```
void reverseString()
```

```
{
```

```
    char str[100];
```

```
    int i, length = 0;
```

```
    printf("Enter a string: ");
```

```
    scanf(" %[^\\n]", str);
```

```
    /* Find length without strlen() */
```

```
    while (str[length] != '\\0')
```

```
    {
```

```
        length++;
```

```
    }
```

```
    printf("Reversed string: ");
```

```
    for (i = length - 1; i >= 0; i--)
```

```
    {
```

```
        printf("%c", str[i]);
```

```
    }
```

```
    printf("\\n");
```

```
}
```

```
/* Function to check palindrome */
void checkPalindrome()
{
    char str[100];
    int i, length = 0;
    int palindrome = 1;

    printf("Enter a string: ");
    scanf("%[^\\n]", str);

    /* Find length without strlen() */
    while (str[length] != '\\0')
    {
        length++;
    }

    for (i = 0; i < length / 2; i++)
    {
        if (str[i] != str[length - 1 - i])
        {
            palindrome = 0;
            break;
        }
    }

    if (palindrome == 1)
        printf("The string is a palindrome.\\n");
}
```

```
else
```

```
    printf("The string is not a palindrome.\n");
```

```
}
```

Test Case Table:

test	choice	input	Expected output
1	1	hello world	HELLO WORLD
2	1	computer	COMPUTER
3	2	hello	olleh
4	2	BTECH	HCETB
5	3	madam	String is a palindrome
6	3	level	String is a palindrome
7	3	hello	String is not a palindrome
8	4	--	Program exits